

2303A51204

Batch:04

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING																		
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026																	
Course Coordinator Name		Dr. Rishabh Mittal																		
Instructor(s) Name		<table border="1"> <tr><td>Mr. S Naresh Kumar</td></tr> <tr><td>Ms. B. Swathi</td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td></tr> <tr><td>Mr. Md Sallauddin</td></tr> <tr><td>Dr. Mathivanan</td></tr> <tr><td>Mr. Y Srikanth</td></tr> <tr><td>Ms. N Shilpa</td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td></tr> <tr><td>Dr. R. Prashant Kumar</td></tr> <tr><td>Mr. Ankushavali MD</td></tr> <tr><td>Mr. B Viswanath</td></tr> <tr><td>Ms. Sujitha Reddy</td></tr> <tr><td>Ms. A. Anitha</td></tr> <tr><td>Ms. M.Madhuri</td></tr> <tr><td>Ms. Katherashala Swetha</td></tr> <tr><td>Ms. Velpula sumalatha</td></tr> <tr><td>Mr. Bingi Raju</td></tr> </table>		Mr. S Naresh Kumar	Ms. B. Swathi	Dr. Sasanko Shekhar Gantayat	Mr. Md Sallauddin	Dr. Mathivanan	Mr. Y Srikanth	Ms. N Shilpa	Dr. Rishabh Mittal (Coordinator)	Dr. R. Prashant Kumar	Mr. Ankushavali MD	Mr. B Viswanath	Ms. Sujitha Reddy	Ms. A. Anitha	Ms. M.Madhuri	Ms. Katherashala Swetha	Ms. Velpula sumalatha	Mr. Bingi Raju
Mr. S Naresh Kumar																				
Ms. B. Swathi																				
Dr. Sasanko Shekhar Gantayat																				
Mr. Md Sallauddin																				
Dr. Mathivanan																				
Mr. Y Srikanth																				
Ms. N Shilpa																				
Dr. Rishabh Mittal (Coordinator)																				
Dr. R. Prashant Kumar																				
Mr. Ankushavali MD																				
Mr. B Viswanath																				
Ms. Sujitha Reddy																				
Ms. A. Anitha																				
Ms. M.Madhuri																				
Ms. Katherashala Swetha																				
Ms. Velpula sumalatha																				
Mr. Bingi Raju																				
CourseCode	23CS002PC304	Course Title	AI Assisted Coding																	
Year/Sem	III/II	Regulation	R23																	
Date and Day of Assignment	Week2 –	Time(s)	23CSBTB01 To 23CSBTB52																	
Duration	2 Hours	Applicable to Batches	All batches																	
Assignment Number: 4.1 (Present assignment number)/ 24 (Total number of assignments)																				
Q.No.	Question	Expected Time to complete																		
1	Question 1: Zero-Shot Prompting (Leap Year Check)	Week2 -																		

Write a zero-shot prompt to generate a Python function that checks whether a given year is a leap year.

Task:

Record the AI-generated code.

Test with years like 1900, 2000, 2024.

Identify logical flaws or missing conditions.

Prompt:

Write a zero-shot prompt to generate a Python function that checks whether a given year is a leap year

Code:

```
def is_leap_year(year):  
    """  
    Function to check whether a given year is a leap year.  
    """  
    if year % 400 == 0:  
        return True  
    elif year % 100 == 0:  
        return False  
    elif year % 4 == 0:  
        return True  
    else:  
        return False  
  
# Test cases  
years = [1900, 2000, 2024]  
  
for y in years:  
    if is_leap_year(y):  
        print(f"{y} is a Leap Year")  
    else:  
        print(f"{y} is NOT a Leap Year")
```

Output:

```
PS C:\Users\madha\OneDrive\Desktop\full_stack_2026> & C:/Users/madha/AppData/Local/Microsoft/Windows/PowerShell/PowerShell.exe  
1900 is NOT a Leap Year  
2000 is a Leap Year  
2024 is a Leap Year  
PS C:\Users\madha\OneDrive\Desktop\full_stack_2026>
```

Explanation

- This is a zero-shot prompt because no examples are given.
- The AI must rely only on its general knowledge of leap year rules.

- Leap year logic:
 - Divisible by 4 ✓
 - Not divisible by 100 ✗
 - Unless divisible by 400 ✓

Question 2: One-Shot Prompting (GCD of Two Numbers)

Write a one-shot prompt with one example to generate a Python function that finds the Greatest Common Divisor (GCD) of two numbers.

Example:

Input: 12, 18 → Output: 6

Task:

Compare with a zero-shot solution.

Analyze algorithm efficiency.

Prompt

Write a Python function to find the Greatest Common Divisor (GCD) of two numbers.

Code:

```
def gcd_zero_shot(a, b):
    gcd = 1
    for i in range(1, min(a, b) + 1):
        if a % i == 0 and b % i == 0:
            gcd = i
    return gcd

def gcd_one_shot(a, b):
    while b != 0:
        a, b = b, a % b
    return a

# Test input
x = 12
y = 18

print("Zero-Shot GCD Result:", gcd_zero_shot(x, y))
print("One-Shot GCD Result:", gcd_one_shot(x, y))
```

Output:

```
PS C:\Users\madha\OneDrive\Desktop\full_stack_2026> .\C:\Users\madha\AppData\Local\Microsoft\Windows\Temporary Internet Files\Content.IE5\QJ5GQJ5G\full_stack_2026\Zero-Shot GCD Result: 6
One-Shot GCD Result: 6
PS C:\Users\madha\OneDrive\Desktop\full_stack_2026>
```

```
PS C:\Users\madha\OneDrive\Desktop\full_stack_2026> & C:/Users/madha/AppData/Local/Microsoft/WindowsApps/python.exe -c "import sys; print(sys.argv[1]);" Zero-Shot GCD Result: 6
One-Shot GCD Result: 6
PS C:\Users\madha\OneDrive\Desktop\full_stack_2026>
```

Explanation

- This is **one-shot prompting** because **one example** is provided.
- The example helps the AI infer:
 - Expected input format
 - Correct output

Question 3: Few-Shot Prompting (LCM Calculation)

Write a few-shot prompt with multiple examples to generate a Python function that computes the Least Common Multiple (LCM).

Examples:

Input: 4, 6 $\rightarrow$ Output: 12

Input: 5, 10 → Output: 10

Input: 7, 3 $\rightarrow$ Output: 21

Task:

Examine how examples guide formula selection.

Test edge cases.

Prompt

Write a Python function to compute the Least Common Multiple (LCM) of two numbers.

Code:

```
def gcd(x, y):
    while y != 0:
        x, y = y, x % y
    return x
def lcm(a, b):
    return (a * b) // gcd(a, b)
# Test cases from the prompt
test_cases = [(4, 6), (5, 10), (7, 3)]

print("LCM Results:")
for a, b in test_cases:
    print(f"LCM of {a} and {b} is {lcm(a, b)}")
# Edge case testing
print("\nEdge Case Testing:")
edge_cases = [(0, 5), (5, 0), (1, 1), (9, 9)]

for a, b in edge_cases:
    print(f"LCM of {a} and {b} is {lcm(a, b)}")
```

Output:

```
LCM Results:
LCM of 4 and 6 is 12
LCM of 5 and 10 is 10
LCM of 7 and 3 is 21

Edge Case Testing:
LCM of 0 and 5 is 0
LCM of 5 and 0 is 0
LCM of 1 and 1 is 1
LCM of 9 and 9 is 9
```

Explanation

- This is **few-shot prompting** because **multiple examples** are given.
- The AI clearly identifies the pattern:
 - $\text{LCM}(a, b) = (a \times b) // \text{GCD}(a, b)$

Question 4: Zero-Shot Prompting (Binary to Decimal Conversion)

Write a zero-shot prompt to generate a Python function that converts a

binary number to decimal.

Task:

Test with valid and invalid binary inputs.

Identify missing validation logic.

Prompt

Write a Python function that converts a binary number to its decimal equivalent.

Code:

```
def binary_to_decimal(binary):  
    return int(binary, 2)  
test_inputs = ["1010", "1111", "1001", "1021", "abc", ""]  
  
print("Binary to Decimal Conversion Results:")  
for b in test_inputs:  
    try:  
        result = binary_to_decimal(b)  
        print(f"Binary: {b} → Decimal: {result}")  
    except ValueError:  
        print(f"Binary: {b} → Invalid binary input")
```

Output:

```
Binary to Decimal Conversion Results:  
Binary: 1010 → Decimal: 10  
Binary: 1111 → Decimal: 15  
Binary: 1001 → Decimal: 9  
Binary: 1021 → Invalid binary input  
Binary: abc → Invalid binary input  
Binary:  → Invalid binary input
```

Explanation

- Zero-shot prompt → no examples.
- AI usually:
 - Uses `int(binary, 2)`
 - Or manual base-2 calculation

Question 5: One-Shot Prompting (Decimal to Binary Conversion)

Write a one-shot prompt with an example to generate a Python function that converts a decimal number to binary.

Example:

Input: 10 → Output: 1010

Task:

Compare clarity with zero-shot output.

Analyze handling of zero and negative numbers.

Prompt

Write a Python function that converts a decimal number to binary.

Code:

```
def decimal_to_binary(n):
    if n == 0:
        return "0"

    is_negative = n < 0
    n = abs(n)

    binary = ""
    while n > 0:
        binary = str(n % 2) + binary
        n = n // 2

    if is_negative:
        binary = "-" + binary

    return binary

test_values = [10, 0, -10, 7, 1]

print("Decimal to Binary Conversion Results:")
for val in test_values:
    print(f"Decimal: {val} → Binary: {decimal_to_binary(val)}")
```

Output:

```
Decimal to Binary Conversion Results:
Decimal: 10 → Binary: 1010
Decimal: 0 → Binary: 0
Decimal: -10 → Binary: -1010
Decimal: 7 → Binary: 111
Decimal: 1 → Binary: 1
```

Explanation

- One example clarifies:

	○ Output format (no 0b prefix) • AI often uses: ○ <code>bin(n)[2:]</code> ○ Or repeated division method	
	Question 6: Few-Shot Prompting (Harshad Number Check) Write a few-shot prompt to generate a Python function that checks whether a number is a Harshad (Niven) number. Examples: Input: 18 → Output: Harshad Number Input: 21 → Output: Harshad Number Input: 19 → Output: Not a Harshad Number Task: Test boundary conditions. Evaluate robustness Prompt Write a Python function to check whether a number is a Harshad (Niven) number. Code:	


```
def is_harshad(n):
    if n <= 0:
        return "Not a Harshad Number"

    digit_sum = sum(int(d) for d in str(n))

    if digit_sum == 0:
        return "Not a Harshad Number"

    if n % digit_sum == 0:
        return "Harshad Number"
    else:
        return "Not a Harshad Number"

test_cases = [18, 21, 19]

print("Harshad Number Test Results:")
for num in test_cases:
    print(f"Input: {num} → Output: {is_harshad(num)}")
print("\nBoundary / Edge Case Testing:")
edge_cases = [1, 0, -18, 10, 100, 11]

for num in edge_cases:
    print(f"Input: {num} → Output: {is_harshad(num)}")
```

Output:

```
Harshad Number Test Results:
Input: 18 → Output: Harshad Number
Input: 21 → Output: Harshad Number
Input: 19 → Output: Not a Harshad Number

Boundary / Edge Case Testing:
Input: 1 → Output: Harshad Number
Input: 0 → Output: Not a Harshad Number
Input: -18 → Output: Not a Harshad Number
Input: 10 → Output: Harshad Number
Input: 100 → Output: Harshad Number
Input: 11 → Output: Not a Harshad Number
```

Explanation

- Few-shot prompting helps AI understand:
 - Harshad number definition
 - Output wording

--	--	--